

ML QB UNIT 5

1. Explain the concept of unsupervised learning with examples.

1. Unsupervised learning is a type of machine learning where data does not contain labeled outputs. The algorithm tries to find hidden patterns and relationships from the given data.
2. It works by grouping similar data points together based on their features. This method is mainly used for clustering and pattern discovery.
3. In unsupervised learning, the system learns without predefined categories or answers. The model independently analyzes the structure of the dataset.
4. Clustering is one of the most common unsupervised learning techniques. It groups similar objects into clusters using similarity or distance measures.
5. Customer segmentation is a popular example of unsupervised learning. Customers with similar buying behavior are grouped into the same cluster.
6. Another example is document grouping in text analysis applications. Similar documents are automatically organized based on their content.
7. Unsupervised learning is also used in anomaly detection systems. It helps identify unusual activities or rare patterns in data.
8. Common unsupervised algorithms include K-Means, Hierarchical Clustering, and DBSCAN. These algorithms help discover meaningful insights from unlabeled data.

2. Differentiate between supervised and unsupervised learning.

Feature	Supervised Learning	Unsupervised Learning
Data Type	Uses labeled data with outputs. The correct answer is already known.	Uses unlabeled data without outputs. The algorithm finds hidden patterns itself.
Goal	Predicts outputs or classes from input data. It learns mapping between input and output.	Groups similar data or discovers relationships. It focuses on pattern identification.
Training	Training is done using known target values. The model compares predictions with actual outputs.	Training is done without target labels. The model works only on input features.
Main Tasks	Classification and regression are common tasks. Examples include spam detection and price prediction.	Clustering and association are common tasks. Examples include customer segmentation and grouping.

Feature	Supervised Learning	Unsupervised Learning
Algorithms	Algorithms include Decision Tree, SVM, and Linear Regression. These models require labeled datasets.	Algorithms include K-Means, DBSCAN, and Hierarchical Clustering. These models work on unlabeled datasets.
Accuracy Check	Accuracy can be measured using known outputs. Performance evaluation is easier.	Accuracy is harder to measure because outputs are unknown. Cluster quality measures are often used.
Example	Predicting student marks from study hours is supervised learning. The output is already known during training.	Grouping customers based on purchasing behavior is unsupervised learning. No predefined groups are given.

3. Explain the need for clustering in machine learning.

1. Clustering is needed to group similar data points into meaningful clusters. It helps discover hidden patterns in large datasets.
2. It is useful when data does not contain predefined labels or categories. Clustering automatically organizes data based on similarity measures.
3. Businesses use clustering for customer segmentation and marketing analysis. Customers with similar behavior can be targeted together.
4. Clustering helps in anomaly or fraud detection systems. Unusual data points can be identified as outliers or noise.
5. It is also used in document and image organization applications. Similar files or images are grouped together automatically.
6. Clustering improves data understanding and exploratory analysis. It helps researchers identify relationships within data.
7. Distance measures like Euclidean distance and cosine similarity are used in clustering. These measures determine how close or similar points are.
8. Common clustering algorithms include K-Means, Hierarchical Clustering, and DBSCAN. Each algorithm is suitable for different types of datasets.

4. Explain Euclidean distance and Manhattan distance with examples.

1. Euclidean distance is the straight-line distance between two points in space. It is one of the most widely used distance measures in clustering.

2. The formula for Euclidean distance is:

$$d(p, q) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}$$

3. For example, the distance between points (2,3) and (5,7) is 5. This method is useful for dense and low-dimensional datasets.
4. Manhattan distance measures the total absolute distance between two points. It is also called city block distance or L1 distance.

5. The formula for Manhattan distance is:

$$d(p, q) = \sum_{i=1}^n |p_i - q_i|$$

6. For points (2,3) and (5,7), Manhattan distance is 7. It measures movement only in horizontal and vertical directions.
7. Euclidean distance is sensitive to outliers because it uses squared differences. Manhattan distance is less sensitive and works better with high-dimensional data.
8. Euclidean distance is commonly used in K-Means and KNN algorithms. Manhattan distance is preferred for sparse or grid-based datasets.

5. Explain cosine similarity and its significance in clustering.

1. Cosine similarity measures the similarity between two vectors based on their orientation. It focuses on the angle between vectors instead of actual distance.
2. A higher cosine value indicates greater similarity between data points. The value ranges from -1 to 1.

3. The formula for cosine similarity is:

$$\text{Similarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|}$$

4. A value near 1 means vectors are very similar in direction. A value near 0 means the vectors are unrelated or orthogonal.
5. Cosine similarity ignores magnitude and focuses only on direction. This makes it useful when document length or size is not important.
6. It is widely used in text mining and Natural Language Processing applications. Documents with similar words can be grouped together effectively.
7. Cosine similarity performs well on high-dimensional datasets. It is commonly used in recommendation systems and document clustering.

6. Differentiate between Euclidean distance, Manhattan distance, and cosine similarity.

Feature	Euclidean Distance	Manhattan Distance	Cosine Similarity
Type	It is a distance metric used in clustering. It measures straight-line distance between points.	It is also a distance metric used in machine learning. It measures total absolute distance between coordinates.	It is a similarity measure instead of distance. It measures the angle between vectors.
Formula Basis	It uses square root of squared differences. Distance is calculated geometrically.	It uses sum of absolute differences. Distance is calculated along grid paths.	It uses cosine of angle between vectors. It checks directional similarity.
Focus	It focuses on actual distance between points. Magnitude is very important here.	It focuses on movement along dimensions. Magnitude also affects the result.	It focuses on orientation rather than magnitude. Vector direction is more important.
Range	Its range is from 0 to infinity. Smaller values mean closer points.	Its range is also from 0 to infinity. Lower values indicate higher similarity.	Its range is from -1 to 1. Higher values indicate greater similarity.
Best Use	Best for dense and low-dimensional data. Commonly used in K-Means and KNN.	Best for high-dimensional and sparse datasets. Useful in grid-based movement problems.	Best for text mining and NLP applications. Useful when magnitude does not matter.
Outlier Effect	It is highly sensitive to outliers. Large values can affect clustering quality.	It is less sensitive to outliers. It performs better with noisy data.	It is not directly affected by outliers. Direction matters more than distance.
Example Use	Used in physical distance calculations. Helpful in sensor-based datasets.	Used in city block navigation problems. Suitable for categorical data analysis.	Used in document similarity and recommendation systems. Common in text clustering tasks.

7. Explain K-Means clustering algorithm.

1. K-Means is an unsupervised machine learning algorithm used for clustering data points. It divides data into K distinct and non-overlapping clusters.
2. In this algorithm, each cluster is represented by a centroid or center point. The centroid is calculated as the mean of all points in the cluster.
3. The value of K represents the number of clusters required in the dataset. This value is provided by the user before training begins.
4. First, K random data points are selected as initial centroids. These centroids act as temporary cluster centers.
5. Each data point is assigned to the nearest centroid using Euclidean distance. Similar points are grouped into the same cluster.
6. After assignment, new centroids are calculated by finding the mean of cluster points. The centroid position changes after every iteration.
7. The process of assigning points and updating centroids repeats continuously. Iterations stop when centroids or cluster assignments no longer change.
8. K-Means is fast, simple, and scalable for large datasets. It is widely used in customer segmentation, image segmentation, and data analysis.

8. Apply K-Means clustering on a dataset and interpret the results.

1. Consider a dataset with points such as (185,72), (170,56), and (188,77). Let the number of clusters be $K = 2$.
2. The first two data points are selected as initial centroids. These centroids are $c_1 = (185,72)$ and $c_2 = (170,56)$.
3. Euclidean distance is calculated between each point and both centroids. Each point is assigned to the nearest centroid cluster.
4. After assigning points, new centroids are calculated using the mean of cluster points. The centroid values change after the first iteration.
5. The assignment and centroid update process is repeated again. This iterative process continues until clusters stop changing.
6. In the second iteration, the cluster assignments remain the same. This indicates that the algorithm has converged successfully.
7. The final clusters contain data points that are more similar to each other. Intracluster distance becomes small while intercluster distance becomes high.
8. The result shows that K-Means effectively groups similar observations together. It helps discover hidden structures and patterns in data.

9. Analyze how to find optimal number of clusters in k means clustering.

1. Choosing the correct number of clusters is one of the biggest challenges in K-Means clustering. A poor choice of K can produce inaccurate clusters.
2. Different values of K are tested to identify the most suitable cluster count. The clustering performance is evaluated for each K value.
3. WCSS (Within-Cluster Sum of Squares) is commonly used for evaluation. Lower WCSS values indicate tighter and better-defined clusters.
4. The Elbow Method helps identify the optimal K using a graph. The best value is selected at the elbow point where WCSS reduction slows down.
5. Silhouette Score is another method for finding the optimal number of clusters. It measures how well points fit within their clusters.
6. A silhouette score close to +1 indicates well-separated and dense clusters. Scores near 0 indicate overlapping or unclear clusters.
7. The optimal K should balance cluster compactness and separation. Too many clusters may lead to overfitting while too few reduce accuracy.
8. Proper cluster selection improves clustering quality and interpretability. It helps discover meaningful patterns from the dataset.

10. Explain the Elbow method for selecting optimal number of clusters in k means clustering.

1. The Elbow Method is used to determine the optimal number of clusters in K-Means clustering. It is a simple and effective evaluation technique.
2. The method works by calculating WCSS values for different values of K. WCSS measures the compactness of clusters.
3. As the number of clusters increases, the WCSS value decreases continuously. This happens because clusters become smaller and tighter.
4. Initially, adding more clusters causes a large decrease in WCSS. After a certain point, the decrease becomes much slower.
5. The point where the graph forms an elbow-like bend is called the elbow point. This point represents the optimal number of clusters.
6. To apply the method, K-Means is executed for multiple K values. The WCSS values are then plotted on a graph.
7. The elbow point provides a balance between accuracy and overfitting. It avoids using too many unnecessary clusters.
8. The Elbow Method is widely used because it is easy to understand and implement. It helps select suitable clusters for effective data analysis.

11. Explain the Silhouette method for selecting optimal number of clusters in k means clustering.

1. The Silhouette method is used to evaluate the quality of clustering results. It helps determine the optimal number of clusters in K-Means clustering.
2. The silhouette score measures how similar a data point is to its own cluster. It also checks how different the point is from other clusters.
3. The score ranges from -1 to +1. A value close to +1 indicates well-separated and compact clusters.
4. A score near 0 means clusters overlap or points are near boundaries. Negative values indicate that points may belong to the wrong cluster.
5. Two measures are used in silhouette calculation called intra-cluster distance and nearest-cluster distance. These values help measure cohesion and separation.
6. The average silhouette score is calculated for different values of K. The K value with the highest score is considered optimal.
7. The method provides both numerical and visual understanding of cluster quality. Silhouette plots help analyze cluster separation clearly.
8. The Silhouette method is widely used because it works without true labels. It is effective for evaluating unsupervised clustering models.

12. Differentiate between Elbow method and Silhouette method for selecting optimal number of clusters in k means clustering.

Feature	Elbow Method	Silhouette Method
Purpose	It finds the optimal number of clusters using WCSS values. It identifies the elbow point in a graph.	It evaluates clustering quality using silhouette scores. It selects the K value with the highest score.
Working Principle	It measures how WCSS decreases as K increases. The bend in the graph indicates optimal K.	It measures cohesion and separation between clusters. Higher scores indicate better clustering quality.
Evaluation Metric	Uses Within-Cluster Sum of Squares (WCSS). Lower WCSS indicates tighter clusters.	Uses silhouette score ranging from -1 to +1. Higher values indicate better-separated clusters.

Feature	Elbow Method	Silhouette Method
Graph Interpretation	The elbow point is visually selected from the graph. Interpretation may sometimes be subjective.	The highest silhouette score determines the best K value. It gives clearer numerical evaluation.
Complexity	It is simple and computationally efficient. It is easier to implement on large datasets.	It is computationally expensive because pairwise distances are calculated. It may take more processing time.
Accuracy	It may not always clearly show the elbow point. Results can sometimes be confusing.	It provides better cluster quality evaluation. It gives more reliable results for separation analysis.
Main Focus	Focuses mainly on cluster compactness. It reduces WCSS while avoiding overfitting.	Focuses on both compactness and separation. It ensures well-defined clusters.

13. Analyze the advantages and limitations of K-Means clustering.

1. K-Means clustering is computationally efficient and works very fast on large datasets. Its simple logic makes implementation easy and understandable.
2. The algorithm scales well with large and high-dimensional datasets. It is widely used in customer segmentation and image analysis applications.
3. K-Means produces clear and interpretable clustering results. The clusters formed are easy to visualize and analyze.
4. The algorithm is guaranteed to converge after multiple iterations. It continuously minimizes WCSS until stable clusters are formed.
5. One major limitation is that the number of clusters K must be chosen in advance. Selecting the wrong K can produce poor clustering results.
6. K-Means is highly sensitive to centroid initialization and outliers. Different initial centroids may lead to different final clusters.
7. The algorithm assumes clusters are spherical and similar in size. It struggles with irregular or non-linear shaped clusters.
8. K-Means also depends heavily on data scaling and normalization. Features with larger values can dominate distance calculations.

14. Analyze how initialization of centroids affects K-Means performance.

1. Centroid initialization strongly affects the quality of K-Means clustering results. Poor initialization may produce inaccurate or unstable clusters.
2. K-Means converges to a local minimum instead of a guaranteed global optimum. Different initial centroids can lead to different clustering outputs.
3. Poor centroid placement may increase the number of iterations required for convergence. This reduces computational efficiency and increases execution time.
4. Effective initialization reduces WCSS and improves cluster compactness. Better centroids create more accurate and meaningful clusters.
5. Random initialization is simple but may place centroids too close together. This can result in poor cluster formation or empty clusters.
6. K-Means++ is a popular initialization technique that selects centroids far apart from each other. It improves both speed and clustering accuracy.
7. Good initialization increases consistency and stability across multiple runs. Similar results are obtained even after repeated executions.
8. Proper centroid initialization helps K-Means discover natural groupings in data effectively. It improves overall clustering performance and reliability.

15. Explain hierarchical clustering with agglomerative approach.

1. Hierarchical clustering is an unsupervised learning technique used to group similar data points. It creates a hierarchy or tree-like structure of clusters.
2. Agglomerative clustering is a bottom-up hierarchical clustering approach. It starts with each data point as an individual cluster.
3. The algorithm calculates distances between all clusters using distance measures. The two closest clusters are merged together first.
4. After merging, the distance matrix is updated again for the remaining clusters. The process continues iteratively until one cluster remains.
5. A dendrogram is used to visualize the merging process of clusters. It shows relationships and similarity between data groups.
6. Lower merge heights in the dendrogram indicate higher similarity between clusters. Cutting the dendrogram at a certain height gives the required clusters.
7. Different linkage methods are used in agglomerative clustering such as single, complete, average, and Ward linkage. These methods determine how cluster distances are calculated.
8. Agglomerative clustering is simple and useful for small to medium datasets. It is widely used in customer segmentation, bioinformatics, and document clustering.

16. Explain hierarchical clustering with divisive approach.

1. Divisive clustering is a top-down hierarchical clustering approach in machine learning. It starts with all data points grouped into one large cluster.
2. The algorithm recursively divides the large cluster into smaller clusters. Splitting continues until each data point becomes its own cluster.
3. In divisive clustering, the most dissimilar points are identified first. These points are used to split the cluster into separate groups.
4. After the first split, the same process is applied to the new clusters again. The algorithm repeatedly divides clusters step by step.
5. The approach creates a hierarchy of clusters that can be represented using a dendrogram. This helps visualize relationships between data groups.
6. Divisive clustering is useful when data naturally forms hierarchical structures. It helps understand cluster organization at multiple levels.
7. This method is computationally expensive because repeated splitting is required. The performance also depends on the splitting strategy used.
8. Divisive clustering is commonly used in taxonomy, document grouping, and biological data analysis. It helps identify meaningful subgroup structures in datasets.

17. Explain dendrogram and its role in hierarchical clustering.

1. A dendrogram is a tree-like diagram used in hierarchical clustering. It visually represents the arrangement and relationships between clusters.
2. Each level in the dendrogram represents a merge or split operation. The height of branches shows the distance between clusters.
3. At the bottom of the dendrogram, each data point appears as an individual cluster. As we move upward, similar clusters are gradually combined.
4. Lower merge heights indicate higher similarity between clusters. Higher merge points indicate that clusters are less similar.
5. Dendrograms help users visually inspect cluster formation processes. They provide a clear understanding of hierarchical relationships in data.
6. The number of clusters can be selected by cutting the dendrogram at a specific height. Different cut levels produce different cluster groupings.
7. Dendrograms are important for exploratory data analysis and pattern discovery. They help identify natural structures in datasets.
8. Hierarchical clustering heavily depends on dendrogram visualization for interpretation. It makes clustering results easier to understand and analyze.

18. Analyze how clusters are formed using hierarchical clustering.

1. Hierarchical clustering forms clusters by continuously merging or splitting data groups. The process depends on similarity or distance measures between points.
2. In agglomerative clustering, each data point initially acts as its own cluster. The closest clusters are merged together step by step.
3. Distances between clusters are calculated using linkage methods. Common methods include single, complete, average, and Ward linkage.
4. After merging two clusters, the distance matrix is updated again. The nearest clusters are then merged in the next iteration.
5. In divisive clustering, all data points begin in one large cluster. The algorithm repeatedly splits clusters into smaller groups.
6. The process continues until stopping conditions are reached or individual clusters remain. This creates a hierarchy of clusters in the dataset.
7. A dendrogram visually shows how clusters are formed during the process. Lower merge heights represent more similar data points.
8. Hierarchical clustering forms meaningful clusters without requiring predefined cluster numbers. It is useful for discovering natural relationships in data.

19. Compare K-Means and hierarchical clustering.

Feature	K-Means Clustering	Hierarchical Clustering
Type	It is a partition-based clustering algorithm. Data is divided into K predefined clusters.	It is a hierarchy-based clustering algorithm. Clusters are formed using merging or splitting methods.
Cluster Number	The number of clusters K must be specified before training. Proper K selection is important.	Number of clusters is not required initially. Clusters can be selected later using a dendrogram.
Working Approach	It works iteratively using centroids and distance calculations. Points are assigned to nearest centroids.	It works by merging or splitting clusters step by step. Relationships are represented hierarchically.
Speed	K-Means is computationally fast and scalable. It performs well on large datasets.	Hierarchical clustering is slower and computationally expensive. It is suitable for small or medium datasets.

Feature	K-Means Clustering	Hierarchical Clustering
Cluster Shape	It assumes clusters are spherical and similar in size. It struggles with irregular shapes.	It can identify complex hierarchical relationships in data. It is more flexible for cluster structures.
Output	Produces only final clusters after convergence. No hierarchy information is shown.	Produces a dendrogram showing cluster relationships. The hierarchy can be visualized clearly.
Sensitivity	Sensitive to centroid initialization and outliers. Different runs may produce different results.	Less dependent on initialization because no random centroids are used. Results are more stable.
Applications	Used in customer segmentation and image segmentation. Best for large datasets with known K.	Used in bioinformatics and taxonomy analysis. Best when hierarchical relationships are important.

20. Explain DBSCAN clustering in detail.

1. DBSCAN stands for Density-Based Spatial Clustering of Applications with Noise. It groups closely packed data points into clusters based on density.
2. DBSCAN identifies dense regions separated by low-density regions in data space. It can detect clusters of arbitrary and irregular shapes.
3. The algorithm uses two important parameters called eps and minPoints. Eps defines neighborhood distance while minPoints defines minimum cluster size.
4. If a point has enough neighboring points within eps distance, it becomes a core point. Core points belong to dense regions of the dataset.
5. Border points have fewer neighbors but lie within the neighborhood of a core point. Noise points are isolated points that do not belong to any cluster.
6. DBSCAN starts with an unvisited point and checks its neighborhood density. If it is a core point, a new cluster is created and expanded.
7. The algorithm continues expanding clusters using density-connected points. Unclustered points that do not satisfy conditions remain as noise.
8. DBSCAN is robust to outliers and does not require specifying the number of clusters beforehand. It is widely used in anomaly detection, spatial analysis, and customer segmentation.

21. Explain ϵ (epsilon) and MinPts parameters in DBSCAN.

1. DBSCAN uses two important parameters called ϵ (epsilon) and MinPts. These parameters help identify dense regions in the dataset.
2. Epsilon defines the maximum distance between two points to be considered neighbors. Points within this radius are treated as part of the same neighborhood.
3. A small epsilon value may create many small clusters and noise points. A large epsilon value may merge different clusters incorrectly.
4. MinPts represents the minimum number of points required to form a dense region. It determines whether a point becomes a core point or not.
5. If a point has neighbors greater than or equal to MinPts within epsilon distance, it is called a core point. Core points help expand clusters.
6. Border points have fewer than MinPts neighbors but lie near a core point. Noise points do not satisfy either condition.
7. Proper selection of epsilon and MinPts improves clustering quality significantly. Incorrect values may lead to over-segmentation or under-segmentation.
8. These parameters make DBSCAN flexible for handling irregular and noisy datasets. They help identify clusters based on density rather than shape.

22. Analyze how DBSCAN identifies core, border, and noise points.

1. DBSCAN classifies data points into core, border, and noise points using density information. The classification depends on epsilon and MinPts values.
2. A core point has at least MinPts neighboring points within epsilon distance. Core points always belong to dense regions in the dataset.
3. If two core points are within epsilon distance, they are connected using density edges. Density-connected core points form the main structure of clusters.
4. Border points have fewer than MinPts neighbors within epsilon distance. However, they lie inside the neighborhood of a core point.
5. Border points belong to a cluster but cannot expand it further. They are attached to nearby dense regions formed by core points.
6. Noise points are isolated points that are neither core nor border points. They do not belong to any cluster in the dataset.
7. DBSCAN starts by selecting an unvisited point and checking its neighborhood density. Core points create clusters while border points join nearby clusters.
8. This density-based classification helps DBSCAN detect arbitrary-shaped clusters and outliers effectively. It performs well on noisy real-world datasets.

23. Compare DBSCAN with K-Means clustering.

Feature	DBSCAN Clustering	K-Means Clustering
Clustering Type	It is a density-based clustering algorithm. Clusters are formed using dense regions.	It is a centroid-based clustering algorithm. Clusters are formed around centroids.
Number of Clusters	Number of clusters is not required beforehand. Clusters are discovered automatically.	Number of clusters K must be specified before training. Incorrect K affects performance.
Cluster Shape	It can identify arbitrary and irregular shaped clusters. It works well with complex structures.	It assumes clusters are spherical and similar in size. It struggles with irregular shapes.
Outlier Handling	It effectively identifies noise and outliers separately. Noise points are not assigned to clusters.	It is sensitive to outliers because centroids get affected. Outliers may distort cluster quality.
Parameters	Uses epsilon and MinPts for clustering. Proper parameter tuning is important.	Uses K value and centroid initialization. Centroid selection affects final results.
Scalability	Slower for very large datasets with high dimensions. Pairwise density calculations increase complexity.	Faster and more scalable for large datasets. It is computationally efficient.
Cluster Formation	Clusters are formed using density-connected points. Core points help expand clusters.	Clusters are formed using nearest centroid assignment. Euclidean distance is commonly used.
Applications	Used in anomaly detection and spatial analysis. Suitable for noisy and irregular datasets.	Used in customer segmentation and image segmentation. Best for compact and well-defined clusters.

24. Analyze advantages and limitations of DBSCAN.

1. DBSCAN can identify clusters of arbitrary and irregular shapes effectively. It performs well on real-world datasets with complex structures.
2. The algorithm is robust to outliers because isolated points are treated as noise. Noise points are not forced into clusters unnecessarily.
3. DBSCAN does not require specifying the number of clusters beforehand. This makes it flexible for exploratory data analysis.
4. It uses only two main parameters called epsilon and MinPts. Parameter tuning is relatively simple compared to some other algorithms.
5. DBSCAN works effectively for spatial and geographical datasets. It is widely used in anomaly detection and traffic analysis.
6. One limitation is sensitivity to parameter selection. Incorrect epsilon or MinPts values may produce poor clustering results.
7. The algorithm struggles with datasets containing clusters of varying densities. Sparse and dense clusters together can reduce clustering quality.
8. DBSCAN also cannot predict cluster membership for new unseen data points. It is mainly applied only to the existing dataset.

25. Analyze which clustering technique is suitable for different types of datasets.

1. K-Means clustering is suitable for large datasets with spherical and compact clusters. It works efficiently when the number of clusters is known beforehand.
2. K-Means performs best on continuous numerical data with low noise. It is commonly used in customer segmentation and image segmentation tasks.
3. Hierarchical clustering is suitable for small and medium-sized datasets. It is useful when hierarchical relationships between data points are important.
4. Hierarchical methods work well when the number of clusters is unknown initially. Dendrograms help visualize cluster structures clearly.
5. Agglomerative clustering is effective for exploratory analysis and taxonomy problems. It is widely used in bioinformatics and document grouping.
6. DBSCAN is suitable for datasets with irregular and arbitrary-shaped clusters. It performs well on noisy and spatial datasets.
7. DBSCAN is preferred when outlier detection is important in the application. It can identify noise points separately from clusters.
8. The choice of clustering technique depends on dataset size, shape, density, and presence of noise. Proper algorithm selection improves clustering quality and interpretation.

(numericals are not included please do them on your own – Kmeans , DBscan , agglomerative)

